

Alessandro Zito

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HIGHLIGHT

Data science postdoc at Harvard Biostatistics and the Dana-Farber Cancer Institute. My work develops generative stochastic process methods, with specific applications to sequencing data in cancer genomics, ecology, and biodiversity. I also work to develop the foundations of Agentic AI and Bayesian decision theory, with applications to medicine.

Ph.D. graduate in Statistics from Duke University. Great experience in applied and methodological research, with particular interests in discrete data, nonparametric methods, and high-dimensional machine learning.

Extensive research collaborations with scientists and physicians outside the statistical field, and with data scientists in industry (Generali and Google).

In recognition of my research, I received the 2024 Savage Award for Outstanding Dissertation, the Lindley Prize (honorable mention), and the BEST Ph.D. student award in Statistical Science at Duke. I also provide service to the statistical community as a board member of J-ISBA.

EXPERIENCE

Postdoctoral researcher, Biostatistics

Boston, MA, USA

Harvard School of Public Health and Dana-Farber Cancer Institute

Sep 2023 – current

- Working with Prof. [Jeff Miller](#) Prof. [Giovanni Parmigiani](#) and Dr. [Mehmet K. Samur](#) on generative methods for cancer genomics, especially mutational processes in DNA and RNA.
- Research areas: Stochastic processes, high-dimensional statistics, dimensionality reduction, regression methods, cancer genomics, multiverse analysis, agentic AI, decision theory.

Intern Data Scientist

San Bruno, CA, USA

Google LLC, Youtube Data Science team

May 2022 – Aug 2022

- Implemented Bayesian hypothesis testing methods to evaluate platform experiments
- Designed a methodology to construct proxy metrics to predict the long-term outcome of experiments, using multivariate Bayesian optimization algorithms and Gaussian processes.
- Extended the collaboration after the internship to publish the methods. See paper [here](#).

Intern Data Scientist

Milan, Italy

Generali Italia S.p.A., Advanced Analytics Solutions team.

Jun 2018 – Jul 2019

- Developed and implemented a Bayesian bandit algorithm recommendation system in Python via a Google Cloud Platform API to optimize a life insurance policy marketing campaign
- Implemented a drivers profiling model, migrated from the partner society MyDrive Solutions, using Python

EDUCATION

Duke University

Durham, NC, USA

Ph.D. in Statistical Science.

Aug 2019 – Sep 2023

- Supervisor: Professor [David B. Dunson](#). Co-supervisor: Professor [Tommaso Rigon](#)
- Research interests: Bayesian Methods, Bayesian Nonparametrics, Species Sampling Models, Ecology
- Funded by project [Lifeplan](#), a European Research Council grant aiming at mapping global trends in biodiversity.
- Dissertation: *Ecological modeling via Bayesian nonparametric species sampling priors* [Link](#) [Slides](#)
- Receipient of the [2024 Savage Award](#) for Outstanding Dissertation in Applied Methodology

Bocconi University

Milan, Italy

Master's of Science in Economics and Social Sciences

Sep 2016 – Mar 2019

- Graduation grade: 110 cum laude / 110
- Thesis: *Conjugate Fractional Factorial Thompson Sampling via unified skew-normal*, with Prof. [Daniele Durante](#)

Bocconi University

Milan, Italy

Bachelor's of Science in Economics and Social Sciences

Sep 2013 – Oct 2016

- Graduation grade: 110 cum laude / 110
- Exchange program at The University of Chicago, Department of Economics, Fall 2015

AVAILABLE PREPRINTS

- Parmigiani G., **Zito, A.** (2026). Multiverse analysis, Bayesian decision theory and agentic AI: drawing connections. *Invited submission to AStA Advances in Statistical Analysis*
- **Zito, A.**, Rigon, T., Roslin, T., Niittynen, P., Hebert, P. D. N., Zakharov, E.V., , Ratnasingham, S., the iBOL consortium, Ovaskainen, O., and Dunson, D. B. (2026). Predicting global biodiversity via Hubbell regression. [bioRxiv:10.64898/2026.05.29.728852](https://doi.org/10.64898/2026.05.29.728852) *Submitted to Science*
- **Zito, A.**, Parmigiani, G., and Miller, J. W. (2025). Poisson process factorization for mutational signature analysis with genomic covariates. [arXiv:2510.26090](https://arxiv.org/abs/2510.26090), *JASA ACS, major revision*.
- Xue, C., **Zito, A.** and Miller, J. W. (2024), Improved control of Dirichlet location and scale near the boundary , [ArXiv:2410.13050](https://arxiv.org/abs/2410.13050), *Bayesian Analysis, Reject and Resubmit*.
- **Zito, A.** and Miller, J. W. (2024), Compressive Bayesian non-negative matrix factorization for mutational signature analysis, [ArXiv:2404.10974](https://arxiv.org/abs/2404.10974), *Biometrics, minor revision*.

PUBLICATIONS

- Modi A., **Zito, A.**, Parmigiani, G. (2026). Pan-Cancer Genomic Scars of Alternative End Joining and Single-Strand Annealing. [DOI:10.1186/s12885-026-16895-2](https://doi.org/10.1186/s12885-026-16895-2) *BMC Cancer*
- Orsholm, J.*, **Zito, A.***, Somervuo, P., Braga M, Chazlot, N, Roslin, T. and Furneaux, B. (2025). Discovering the unseen: a performance comparison of taxonomic classification methods under unknown DNA barcodes. *Methods in Ecology and Evolution*. [DOI:10.1111/2041-210x.70358](https://doi.org/10.1111/2041-210x.70358)
- Lee, C. J., **Zito, A.**, Sang, H. and Dunson, D. B. (2025), Logistic-beta processes for dependent random probabilities with beta marginals, *Bayesian Analysis*. Advanced Publications 1-25. [DOI:10.1214/25-BA1541](https://doi.org/10.1214/25-BA1541)
- **Zito, A.**, Greaves, D., Soriano, J. and Richardson, L. (2025), Pareto Optimal Proxy Metrics, *Applied Stochastic models in Business and Industry*, 41: e70003. [DOI:10.1002/asmb.70003](https://doi.org/10.1002/asmb.70003)
- **Zito, A.**, Rigon., T. and Dunson, D. B. (2024), Bayesian nonparametric modelling of latent partitions via Stirling-gamma priors, *Bayesian Analysis*. Advanced Publications 1-24. [DOI:10.1214/24-BA1463](https://doi.org/10.1214/24-BA1463).
- **Zito, A.**, Rigon., T., Ovaskainen, O. and Dunson, D. B. (2023), Bayesian modeling of sequential discoveries, *Journal of the American Statistical Association (T&M)*, 118:544, 2521-32, [DOI:10.1080/01621459.2022.2060835](https://doi.org/10.1080/01621459.2022.2060835)
- **Zito, A.**, Rigon., T. and Dunson, D. B. (2023), Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa. *Methods in Ecology and Evolution*. 14, 529–42. [DOI:10.1111/2041-210X.14009](https://doi.org/10.1111/2041-210X.14009)

CONFERENCE PROCEEDINGS AND DISCUSSIONS

- **Zito, A.**, Xue, C., and Miller, J. W. (2025). Accounting for Transcriptional Asymmetries in Mutational Signature Analysis Using Compressive Bayesian Non-negative Matrix Factorization. In: di Bella, E., Gioia, V., Lagazio, C., Zaccarin, S. (eds) *Statistics for Innovation IV. SIS 2025. Italian Statistical Society Series on Advances in Statistics*. Springer, Cham. [Link](#)
- **Zito, A.** and Miller, J. W. (2024). Discussion to “Sparse Bayesian factor analysis when the number of factors is unknown” by Fruhwirth-Schnatter, S., Hosszejni, D., and Lopes, F. L., *Bayesian Analysis*. [publication link]
- **Zito, A.**, Rigon, T. and Dunson, D. B. (2021). Modelling of accumulation curves through Weibull survival functions. In *Book of Short Papers of the Italian Statistical Society 2021* (Perna, C., Salvati, N. and Schirripa Spagnolo, F., editors). ISBN: 9788891927361.

MANUSCRIPTS IN PREPARATION

- **Zito, A.**, Parmigiani, G., Miller, J. W. and Samur, M. (2026). Pan-cancer analysis of RNA mutational signatures.
- Qian Z., Parmigiani, G., Samur, M. and **Zito, A.** (2026). Temporal non-negative matrix factorization for diagnosis-relapse in multiple myeloma
- **Zito, A.** and Parmigiani, G. (2026). Multiverse analysis with AI agents.

SOFTWARE AND WEBSITES

- hubbellglm.org: an online platform to monitor arthropod diversity
- **Zito, A.** (2026) HubbellGLM: Package to run Hubbell regression [GitHub](#)
- **Zito, A.** (2023) CompressiveNMF: Bayesian compressive non-negative matrix factorization software [GitHub](#)
- **Zito, A.** (2023) ConjugateDP: Random number generator for the Stirling-gamma distribution. [GitHub](#)
- **Zito, A.** (2022) BayesANT: Bayesian nonparametric taxonomic classifier for DNA barcoding sequences. [GitHub](#)
- **Zito, A., Rigon, T.** (2021) BNPvegan: Bayesian nonparametric methods for ecology. [GitHub](#)

SKILLS

IT: R, Rcpp (Fluent), Python, Keras, Tensorflow, Git, Cluster computing via SLURM, SQL (Proficient)
Languages: Italian (Native), English (Fluent), Spanish (Proficient)

PROFESSIONAL ACTIVITIES

Board member and treasurer of j-ISBA, the junior section of the International Society for Bayesian Analysis. Current duties include organizing the 2025 and 2026 Blackwell-Rosenbluth awards, organizing BAYSM2026 in Chiba, Japan, and expanding j-ISBA's outreach to lower-income countries.

Referee activity: *Journal of the American Statistical Association* (2), *Annals of Applied Statistics* (3), *Journal of Computational and Graphical Statistics* (1), *Journal of Machine Learning research* (1) *PeerJ* (1), *The American Statistician* (1), *Electronic Journal of Statistics* (2), *Bayesian Analysis* (2), *Computational Statistics & Data Analysis* (1), *Ecography* (1), *Philosophical transactions of the Royal Society A* (1).

AWARDS

Lindley Prize 2025 - Honorable mention
BNP14 Travel Award
ISBA 2024 Savage Award for Outstanding Dissertation in Bayesian Applied Methodology
ISBA 2022 Travel Award
BEST Award for Ph.D. student research for the 2021/2022 academic year
Ph.D. Teaching Assistant of the Year Award for the 2020/2021 academic year
Best Student/Postdoc Contributed Paper Award at ISBA 2021 World meeting
Bocconi Merit Award scholarship for the 2016/2017 and 2017/2018 academic years

ORAL PRESENTATIONS AND TALKS

Sapienza University, Rome, Italy – *Statistical methods for mutational processes in cancer* (Invited talk)
MD Anderson Cancer Center, Houston, TX – *Modeling mutational processes in cancer via Poisson Process Factorization* (Invited talk)
Swarthmore College, Swarthmore, PA – *Statistical methods for mutational processes in cancer* (Invited talk)
JSM2025, Nashville, TN – *Ecological Modeling via Bayesian Nonparametric Species Sampling Priors* (Invited talk, The Savage Dissertation Award)
BNP14, Los Angeles, CA – *Topography-dependent mutational signature analysis via Poisson process factorization* (Contributed talk)
National Center for Statistical Ecology, University of Kent, United Kingdom – *Predicting global Biodiversity via Hubbell regression* (Invited online talk)
ENAR2025, New Orleans, LA – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Contributed talk)
Brown University, Data Science Initiative, Providence, RI, USA – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited seminar)
Cattolica University of the Sacred Heart, Milan, Italy – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited seminar)
ICSDS2024, Nice, France – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Contributed talk)
ISBA2024, Venice, Italy – *Compressive Bayesian non-negative matrix factorization for mutational signature analysis* (Invited talk)
ISBA2024, Venice, Italy – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Poster)

CMstat2023, Berlin, Germany – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)

EcoSta2023, Tokyo, Japan – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)

NESS2023, Boston, MA – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Invited talk)

CMstat2022, London, UK – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Invited talk)

BNP13, Puerto Varas, Chile – *Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors* (Poster)

ISBA2022, Montreal, Canada – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Poster)

Luomos SYKE Machine Learning Seminar II (Virtual) – *Taxonomic classification methods for DNA sequences* (Contributed talk)

ISBA2021 (Virtual) – *Bayesian modeling of sequential discoveries* (Contributed talk)

ISBA2021 (Virtual) – *Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa* (Contributed talk)

SESSIONS ORGANIZED AND CHAIRED

BAYSM2025 and **BAYSM2026**. Member of the scientific and organizing committee.

JSM2025, Nashville, TN – *Advances in Bayesian Factorization Methods in Genomics and Medicine* (Invited session).
Speakers: Adolphus Wagala (Dana-Farber Cancer Institute), Jonathan Huggins (Boston University), Sarah Urbut (Broad Institute), Niccolo Anceschi (Duke University). Discussion by Peter Carbonetto (University of Chicago).

ISBA2024, Venice, Italy – *Bayesian Methods in Ecology* (Invited session).

Speakers: Bokgyeong Kang (Duke University), Michele Peruzzi (University of Michigan), Narmadha M. Mohankumar (Pacific Northwest National Lab).

OTHER ACHIEVEMENTS

Painter. Personal exhibitions at Triennale di Milano museum of arts and “Spazio Arte Tirabassi” art gallery.
Winner of the 2018 “Looking for Art” contest in Milan.
Former semi-professional actor at “Proxima Res”, Milan